

16. 3.2

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

(a) orthon. bázis podpr. $\text{span}\{\vec{n}_1, \vec{n}_2, \vec{n}_3\}$ prostoru V

$$1.) \vec{u}_1 = \frac{1}{\|\vec{n}_1\|} \cdot \vec{n}_1 = \frac{1}{2} \cdot 1 = \boxed{\frac{1}{2}} \quad \text{span}\{\vec{n}_1\} = \text{span}\{\vec{u}_1\}$$

$$\hookrightarrow \sqrt{\int_1^5 1 dx} = \sqrt{4}$$

$$2.) \vec{u}_2 = \frac{1}{\|\vec{n}_2 - \vec{n}_1\|} \cdot (\vec{n}_2 - \vec{n}_1), \text{ kde } \vec{n}_2 = \langle \vec{u}_1, \vec{n}_2 \rangle \cdot \vec{u}_1$$

$$\vec{n}_2 = \left\langle \frac{1}{2}, x \right\rangle \cdot \frac{1}{2} = \int_1^5 \frac{x}{2} dx \cdot \frac{1}{2} = \frac{1}{4} \int_1^5 x dx = \frac{1}{4} \cdot 12 = 3$$

$$\vec{u}_2 = \frac{1}{\|x-3\|} \cdot (x-3) = \frac{1}{\sqrt{\int_1^5 (x-3)^2 dx}} \cdot (x-3) = \frac{1}{\sqrt{\frac{16}{3}}} (x-3) = \frac{\sqrt{3}}{\sqrt{16}} (x-3) = \boxed{\frac{\sqrt{3}}{4} (x-3)}$$

$$3.) \vec{u}_3 = \frac{1}{\|\vec{n}_3 - \vec{n}_1\|} (\vec{n}_3 - \vec{n}_1)$$

$$\vec{n}_3 = \langle \vec{u}_1, \vec{n}_3 \rangle \cdot \vec{u}_1 + \langle \vec{u}_2, \vec{n}_3 \rangle \cdot \vec{u}_2 = \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} + \int_1^5 x^2 \cdot \frac{\sqrt{3}}{4} (x-3) dx \cdot \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{1}{2} \cdot \frac{124}{3} \cdot \frac{1}{2} + \frac{\sqrt{3}}{4} \int_1^5 x^3 - 3x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) = \frac{31}{3} + \frac{3}{16} \cdot 32 (x-3) = 6x - 18 + \frac{31}{3} = 6x - \frac{23}{3}$$

$$\vec{u}_3 = \frac{1}{\|x^2 - 6x + \frac{23}{3}\|} \cdot \left(x^2 - 6x + \frac{23}{3}\right) = \boxed{\frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right)}$$

$$\hookrightarrow \sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)} = \sqrt{\frac{256}{45}} = \frac{16}{3\sqrt{5}}$$

Ordonormalua bāra B pyrerā taldo:

$$B = \left(\underbrace{\frac{1}{2}}_{\vec{u}_1}, \underbrace{\frac{\sqrt{3}}{4}(x-3)}_{\vec{u}_2}, \underbrace{\frac{3-\sqrt{5}}{14}\left(x^2-6x+\frac{23}{3}\right)}_{\vec{u}_3} \right)$$

(b) Máme ON řávn, vyjádříme $\sin x$ v těchto bázích:

$$[\sin x]_B = \begin{pmatrix} \langle \vec{u}_1, \sin x \rangle \\ \langle \vec{u}_2, \sin x \rangle \\ \langle \vec{u}_3, \sin x \rangle \end{pmatrix} = \begin{pmatrix} \langle \frac{1}{2}, \sin x \rangle \\ \langle \frac{\sqrt{3}}{4}(x-3), \sin x \rangle \\ \langle \frac{3\sqrt{5}}{16}(x^2-6x+\frac{23}{3}), \sin x \rangle \end{pmatrix}$$

$$\langle \frac{1}{2}, \sin x \rangle = \int_1^5 \frac{1}{2} \sin x \, dx = \frac{1}{2} \int_1^5 \sin x \, dx = \frac{1}{2} (\cos 1 - \cos 5) \approx (0,2566) \frac{1}{2}$$

$$\langle \frac{\sqrt{3}}{4}(x-3), \sin x \rangle = \frac{\sqrt{3}}{4} \int_1^5 (x-3) \sin x \, dx = \frac{\sqrt{3}}{4} (-\sin 1 + \sin 5 - 2\cos 1 - 2\cos 5) \approx (-3,4483) \frac{\sqrt{3}}{4}$$

$$\langle \frac{3\sqrt{5}}{16}(x^2-6x+\frac{23}{3}), \sin x \rangle = \frac{3\sqrt{5}}{16} \int_1^5 (x^2-6x+\frac{23}{3}) \sin x \, dx = \frac{3\sqrt{5}}{16} (\frac{2}{3}(6(\sin 1 + \sin 5) + \cos 1 - \cos 5)) \approx (-0,2987) \frac{3\sqrt{5}}{16}$$

$$\sin x \approx \frac{1}{2} \cdot \frac{1}{2} \cdot 0,2566 + \frac{\sqrt{3}}{4}(x-3) \cdot \frac{\sqrt{3}}{4} \cdot (-3,4483) +$$

$$+ \frac{3\sqrt{5}}{16}(x^2-6x+\frac{23}{3}) \cdot \frac{3\sqrt{5}}{16} (-0,2987) = 0,0642 + \frac{3}{16}(x-3)(-3,4483) +$$

$$+ \frac{45}{256}(x^2-6x+\frac{23}{3})(-0,2987) = 0,0642 - 0,6466(x-3) - 0,0525(x^2-6x+\frac{23}{3}) =$$

$$= 0,0642 - 0,6466x + 1,9398 - 0,0525x^2 + 0,315x - 0,4025 =$$

$$= -0,0525x^2 - 0,3316x + 1,6015$$

(c) vyjádříme $\|\sin x - (ax^2+bx+c)\|$:

$$\sqrt{\int_1^5 (\sin x - (-0,0525x^2 - 0,3316x + 1,6015))^2 \, dx} = \underline{\underline{0,31879}}$$